



The RISC-V Instruction Set Manual, Volume I

Unprivileged Architecture

Version 20260120: Official Release

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Preamble

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Preface

This document describes the RISC-V unprivileged architecture. It contains the following versions of the RISC-V ISA modules, all of which have been ratified:

Base	Version	Status
RV32I	2.1	Ratified
RV32E	2.0	Ratified
RV64E	2.0	Ratified
RV64I	2.1	Ratified
Extension	Version	Status
Zifencei	2.0	Ratified
Zicsr	2.0	Ratified
Zicntr	2.0	Ratified
Zihintntnl	1.0	Ratified
Zihintpause	2.0	Ratified
Zimop	1.0	Ratified
Zicond	1.0	Ratified
Zilsd	1.0	Ratified
M	2.0	Ratified
Zmmul	1.0	Ratified
A	2.1	Ratified
Zawrs	1.01	Ratified
Zacas	1.0	Ratified
Zabha	1.0	Ratified
Zalasr	1.0	Ratified
RVWMO	2.0	Ratified
Ztso	1.0	Ratified
CMO	1.0	Ratified
F	2.2	Ratified
D	2.2	Ratified
Q	2.2	Ratified
Zfh	1.0	Ratified
Zfhmin	1.0	Ratified
BF16	1.0	Ratified
Zfa	1.0	Ratified
Zfinx	1.0	Ratified
Zdinx	1.0	Ratified
Zhinx	1.0	Ratified
Zhinxmin	1.0	Ratified
C	2.0	Ratified

Base	Version	Status
Zce	1.0	Ratified
Zclsd	1.0	Ratified
B	1.0	Ratified
V	1.0	Ratified
Zbkb	1.0	Ratified
Zbkc	1.0	Ratified
Zbkx	1.0	Ratified
Zk	1.0	Ratified
Zks	1.0	Ratified
Zvbb	1.0	Ratified
Zvbc	1.0	Ratified
Zvkg	1.0	Ratified
Zvkned	1.0	Ratified
Zvknhb	1.0	Ratified
Zvksed	1.0	Ratified
Zvksh	1.0	Ratified
Zvkt	1.0	Ratified
Zicfiss	1.0	Ratified
Zicfilp	1.0	Ratified

The changes in this version of the document include:

- Addition of the Zalasr extension for Load-Acquire/Store-Release operations.

Preface to Document Version 20250508

This document describes the RISC-V unprivileged architecture. It contains the following versions of the RISC-V ISA modules, all of which have been ratified:

Base	Version	Status
RV32I	2.1	Ratified
RV32E	2.0	Ratified
RV64E	2.0	Ratified
RV64I	2.1	Ratified
Extension	Version	Status
Zifencei	2.0	Ratified
Zicsr	2.0	Ratified
Zicntr	2.0	Ratified
Zihintntnl	1.0	Ratified
Zihintpause	2.0	Ratified
Zimop	1.0	Ratified
Zicond	1.0	Ratified

Base	Version	Status
Zilsd	1.0	Ratified
M	2.0	Ratified
Zmmul	1.0	Ratified
A	2.1	Ratified
Zawrs	1.01	Ratified
Zacas	1.0	Ratified
Zabha	1.0	Ratified
RVWMO	2.0	Ratified
Ztso	1.0	Ratified
CMO	1.0	Ratified
F	2.2	Ratified
D	2.2	Ratified
Q	2.2	Ratified
Zfh	1.0	Ratified
Zfhmin	1.0	Ratified
BF16	1.0	Ratified
Zfa	1.0	Ratified
Zfinx	1.0	Ratified
Zdinx	1.0	Ratified
Zhinx	1.0	Ratified
Zhinxmin	1.0	Ratified
C	2.0	Ratified
Zce	1.0	Ratified
Zclsd	1.0	Ratified
B	1.0	Ratified
V	1.0	Ratified
Zbkb	1.0	Ratified
Zbkc	1.0	Ratified
Zbkx	1.0	Ratified
Zk	1.0	Ratified
Zks	1.0	Ratified
Zvbb	1.0	Ratified
Zvbc	1.0	Ratified
Zvkg	1.0	Ratified
Zvkned	1.0	Ratified
Zvknhb	1.0	Ratified
Zvksed	1.0	Ratified
Zvksh	1.0	Ratified
Zvkt	1.0	Ratified

Base	Version	Status
Zicfiss	1.0	Ratified
Zicfilp	1.0	Ratified

The changes in this version of the document include:

- The inclusion of all ratified extensions through May 2025.
- Removal of all unratified material.
- Addition of the BFloat16-precision Floating Point extension.
- Addition of the Zabha extension for Byte and Halfword Atomic Memory Operations.

Preface to Document Version 20240411

This document describes the RISC-V unprivileged architecture. It contains the following versions of the RISC-V ISA modules:

Base	Version	Status
RV32I	2.1	Ratified
RV32E	2.0	Ratified
RV64E	2.0	Ratified
RV64I	2.1	Ratified
Extension	Version	Status
Zifencei	2.0	Ratified
Zicsr	2.0	Ratified
Zicntr	2.0	Ratified
Zihintntnl	1.0	Ratified
Zihintpause	2.0	Ratified
Zimop	1.0	Ratified
Zicond	1.0	Ratified
Zilsd	1.0	Ratified
M	2.0	Ratified
Zmmul	1.0	Ratified
A	2.1	Ratified
Zawrs	1.01	Ratified
Zacas	1.0	Ratified
Zabha	1.0	Ratified
RVWMO	2.0	Ratified
Ztso	1.0	Ratified
CMO	1.0	Ratified
F	2.2	Ratified
D	2.2	Ratified
Q	2.2	Ratified
Zfh	1.0	Ratified

Base	Version	Status
Zfhmin	1.0	Ratified
Zfa	1.0	Ratified
Zfinx	1.0	Ratified
Zdinx	1.0	Ratified
Zhinx	1.0	Ratified
Zhinxmin	1.0	Ratified
C	2.0	Ratified
Zce	1.0	Ratified
Zclsd	1.0	Ratified
B	1.0	Ratified
V	1.0	Ratified
Zbkb	1.0	Ratified
Zbkc	1.0	Ratified
Zbkx	1.0	Ratified
Zk	1.0	Ratified
Zks	1.0	Ratified
Zvbb	1.0	Ratified
Zvbc	1.0	Ratified
Zvkg	1.0	Ratified
Zvkned	1.0	Ratified
Zvknhb	1.0	Ratified
Zvksed	1.0	Ratified
Zvksh	1.0	Ratified
Zvkt	1.0	Ratified
Zicfiss	1.0	Ratified
Zicfilp	1.0	Ratified

The changes in this version of the document include:

- The inclusion of all ratified extensions through February 2025.
- The draft Zam extension has been removed, in favor of the definition of a misaligned atomicity granule PMA.
- The concept of vacant memory regions has been superseded by inaccessible memory or I/O regions.
- The removal of unrated content, including the sketch of the RV128I base ISA.

Preface to Document Version 20191213-Base-Ratified

This document describes the RISC-V unprivileged architecture.

The ISA modules marked **Ratified** have been ratified at this time. The modules marked *Frozen* are not expected to change significantly before being put up for ratification. The modules marked *Draft* are expected to change before ratification.

The document contains the following versions of the RISC-V ISA modules:

Base	Version	Status
RVWMO	2.0	Ratified
RV32I	2.1	Ratified
RV64I	2.1	Ratified
RV32E	1.9	Draft
RV128I	1.7	Draft
Extension	Version	Status
M	2.0	Ratified
A	2.1	Ratified
F	2.2	Ratified
D	2.2	Ratified
Q	2.2	Ratified
C	2.0	Ratified
Counters	2.0	Draft
L	0.0	Draft
B	0.0	Draft
J	0.0	Draft
T	0.0	Draft
P	0.2	Draft
V	0.7	Draft
Zicsr	2.0	Ratified
Zifencei	2.0	Ratified
Zam	0.1	Draft
Ztso	0.1	Frozen

The changes in this version of the document include:

- The A extension, now version 2.1, was ratified by the board in December 2019.
- Defined big-endian ISA variant.
- Moved N extension for user-mode interrupts into Volume II.
- Defined PAUSE hint instruction.

Preface to Document Version 20190608-Base-Ratified

This document describes the RISC-V unprivileged architecture.

The RVWMO memory model has been ratified at this time. The ISA modules marked **Ratified**, have been ratified at this time. The modules marked *Frozen* are not expected to change significantly before being put up for ratification. The modules marked *Draft* are expected to change before ratification.

The document contains the following versions of the RISC-V ISA modules:

Base	Version	Status
RVWMO	2.0	Ratified
RV32I	2.1	Ratified
RV64I	2.1	Ratified
RV32E	1.9	Draft
RV128I	1.7	Draft
Extension	Version	Status
Zifencei	2.0	Ratified
Zicsr	2.0	Ratified
M	2.0	Ratified
A	2.0	Frozen
F	2.2	Ratified
D	2.2	Ratified
Q	2.2	Ratified
C	2.0	Ratified
Ztso	0.1	Frozen
Counters	2.0	Draft
L	0.0	Draft
B	0.0	Draft
J	0.0	Draft
T	0.0	Draft
P	0.2	Draft
V	0.7	Draft
Zam	0.1	Draft

The changes in this version of the document include:

- Moved description to **Ratified** for the ISA modules ratified by the board in early 2019.
- Removed the A extension from ratification.
- Changed document version scheme to avoid confusion with versions of the ISA modules.
- Incremented the version numbers of the base integer ISA to 2.1, reflecting the presence of the ratified RVWMO memory model and exclusion of FENCE.I, counters, and CSR instructions that were in previous base ISA.
- Incremented the version numbers of the F and D extensions to 2.2, reflecting that version 2.1 changed the canonical NaN, and version 2.2 defined the NaN-boxing scheme and changed the definition of the FMIN and FMAX instructions.
- Changed name of document to refer to "unprivileged" instructions as part of move to separate ISA specifications from platform profile mandates.
- Added clearer and more precise definitions of execution environments, harts, traps, and memory accesses.
- Defined instruction-set categories: *standard*, *reserved*, *custom*, *non-standard*, and *non-conforming*.
- Removed text implying operation under alternate endianness, as alternate-endianness operation has not yet been defined for RISC-V.

- Changed description of misaligned load and store behavior. The specification now allows visible misaligned address traps in execution environment interfaces, rather than just mandating invisible handling of misaligned loads and stores in user mode. Also, now allows access-fault exceptions to be reported for misaligned accesses (including atomics) that should not be emulated.
- Moved FENCE.I out of the mandatory base and into a separate extension, with Zifencei ISA name. FENCE.I was removed from the Linux user ABI and is problematic in implementations with large incoherent instruction and data caches. However, it remains the only standard instruction-fetch coherence mechanism.
- Removed prohibitions on using RV32E with other extensions.
- Removed platform-specific mandates that certain encodings produce illegal-instruction exceptions in RV32E and RV64I chapters.
- Counter/timer instructions are now not considered part of the mandatory base ISA, and so CSR instructions were moved into separate chapter and marked as version 2.0, with the unprivileged counters moved into another separate chapter. The counters are not ready for ratification as there are outstanding issues, including counter inaccuracies.
- A CSR-access ordering model has been added.
- Explicitly defined the 16-bit half-precision floating-point format for floating-point instructions in the 2-bit *fmt* field.
- Defined the signed-zero behavior of *FMIN.fmt* and *FMAX.fmt*, and changed their behavior on signaling-NaN inputs to conform to the `minimumNumber` and `maximumNumber` operations in the proposed IEEE 754-201x specification.
- The memory consistency model, RVWMO, has been defined.
- The "Zam" extension, which permits misaligned AMOs and specifies their semantics, has been defined.
- The "Ztso" extension, which enforces a stricter memory consistency model than RVWMO, has been defined.
- Improvements to the description and commentary.
- Defined the term **IALIGN** as shorthand to describe the instruction-address alignment constraint.
- Removed text of **P** extension chapter as now superseded by active task group documents.
- Removed text of **V** extension chapter as now superseded by separate vector extension draft document.

Preface to Document Version 2.2

This is version 2.2 of the document describing the RISC-V user-level architecture. The document contains the following versions of the RISC-V ISA modules:

Base	Version	Draft Frozen?
RV32I	2.0	Y
RV32E	1.9	N
RV64I	2.0	Y
RV128I	1.7	N
Extension	Version	Frozen?
M	2.0	Y
A	2.0	Y
F	2.0	Y
D	2.0	Y

Base	Version	Draft Frozen?
Q	2.0	Y
L	0.0	N
C	2.0	Y
B	0.0	N
J	0.0	N
T	0.0	N
P	0.1	N
V	0.7	N
N	1.1	N

To date, no parts of the standard have been officially ratified by the RISC-V Foundation, but the components labeled "frozen" above are not expected to change during the ratification process beyond resolving ambiguities and holes in the specification.

The major changes in this version of the document include:

- The previous version of this document was released under a Creative Commons Attribution 4.0 International License by the original authors, and this and future versions of this document will be released under the same license.
- Rearranged chapters to put all extensions first in canonical order.
- Improvements to the description and commentary.
- Modified implicit hinting suggestion on **JALR** to support more efficient macro-op fusion of **LUI/JALR** and **AUIPC/JALR** pairs.
- Clarification of constraints on load-reserved/store-conditional sequences.
- A new table of control and status register (CSR) mappings.
- Clarified purpose and behavior of high-order bits of **fcsr**.
- Corrected the description of the **FNMADD.fmt** and **FNMSUB.fmt** instructions, which had suggested the incorrect sign of a zero result.
- Instructions **FMV.S.X** and **FMV.X.S** were renamed to **FMV.W.X** and **FMV.X.W** respectively to be more consistent with their semantics, which did not change. The old names will continue to be supported in the tools.
- Specified behavior of narrower (<FLEN) floating-point values held in wider **f** registers using NaN-boxing model.
- Defined the exception behavior of **FMA(∞ , 0, qNaN)**.
- Added note indicating that the **P** extension might be reworked into an integer packed-SIMD proposal for fixed-point operations using the integer registers.
- A draft proposal of the **V** vector instruction-set extension.
- An early draft proposal of the **N** user-level traps extension.
- An expanded pseudoinstruction listing.
- Removal of the calling convention chapter, which has been superseded by the RISC-V ELF psABI Specification ([RISC-V ELF PsABI Specification, n.d.](#)).
- The **C** extension has been frozen and renumbered version 2.0.

Preface to Document Version 2.1

This is version 2.1 of the document describing the RISC-V user-level architecture. Note the frozen user-level ISA base and extensions **IMAFDQ** version 2.0 have not changed from the previous version of this document ([Waterman et al., 2014](#)), but some specification holes have been fixed and the documentation has been improved. Some changes have been made to the software conventions.

- Numerous additions and improvements to the commentary sections.
- Separate version numbers for each chapter.
- Modification to long instruction encodings >64 bits to avoid moving the *rd* specifier in very long instruction formats.
- CSR instructions are now described in the base integer format where the counter registers are introduced, as opposed to only being introduced later in the floating-point section (and the companion privileged architecture manual).
- The SCALL and SBREAK instructions have been renamed to **ECALL** and **EBREAK**, respectively. Their encoding and functionality are unchanged.
- Clarification of floating-point NaN handling, and a new canonical NaN value.
- Clarification of values returned by floating-point to integer conversions that overflow.
- Clarification of **LR/SC** allowed successes and required failures, including use of compressed instructions in the sequence.
- A new **RV32E** base ISA proposal for reduced integer register counts, supports **MAC** extensions.
- A revised calling convention.
- Relaxed stack alignment for soft-float calling convention, and description of the RV32E calling convention.
- A revised proposal for the **C** compressed extension, version 1.9 .

Preface to Version 2.0

This is the second release of the user ISA specification, and we intend the specification of the base user ISA plus general extensions (i.e., **IMAFD**) to remain fixed for future development. The following changes have been made since Version 1.0 ([Waterman et al., 2011](#)) of this ISA specification.

- The ISA has been divided into an integer base with several standard extensions.
- The instruction formats have been rearranged to make immediate encoding more efficient.
- The base ISA has been defined to have a little-endian memory system, with big-endian or bi-endian as non-standard variants.
- Load-Reserved/Store-Conditional (**LR/SC**) instructions have been added in the atomic instruction extension.
- **AMOs** and **LR/SC** can support the release consistency model.
- The **FENCE** instruction provides finer-grain memory and I/O orderings.
- An **AMO** for fetch-and-XOR (**AMOXOR**) has been added, and the encoding for **AMOSWAP** has been changed to make room.
- The **AUIPC** instruction, which adds a 20-bit upper immediate to the **PC**, replaces the **RDNPC** instruction, which only read the current **PC** value. This results in significant savings for position-independent code.
- The **JAL** instruction has now moved to the **U-Type** format with an explicit destination register, and the **J** instruction has been dropped being replaced by **JAL** with *rd*=x0. This removes the only instruction with

an implicit destination register and removes the **J-Type** instruction format from the base ISA. There is an accompanying reduction in **JAL** reach, but a significant reduction in base ISA complexity.

- The static hints on the **JALR** instruction have been dropped. The hints are redundant with the *rd* and *rs1* register specifiers for code compliant with the standard calling convention.
- The **JALR** instruction now clears the lowest bit of the calculated target address, to simplify hardware and to allow auxiliary information to be stored in function pointers.
- The **MFTX.S** and **MFTX.D** instructions have been renamed to **FMV.X.S** and **FMV.X.D**, respectively. Similarly, **MXTF.S** and **MXTF.D** instructions have been renamed to **FMV.S.X** and **FMV.D.X**, respectively.
- The **MFFSR** and **MTFSR** instructions have been renamed to **FRCSR** and **FSCSR**, respectively. **FRRM**, **FSRM**, **FRFLAGS**, and **FSFLAGS** instructions have been added to individually access the rounding mode and exception flags subfields of the **fcsr**.
- The **FMV.X.S** and **FMV.X.D** instructions now source their operands from *rs1*, instead of *rs2*. This change simplifies datapath design.
- **FCLASS.S** and **FCLASS.D** floating-point classify instructions have been added.
- A simpler NaN generation and propagation scheme has been adopted.
- For **RV32I**, the system performance counters have been extended to 64-bits wide, with separate read access to the upper and lower 32 bits.
- Canonical **NOP** and **MV** encodings have been defined.
- Standard instruction-length encodings have been defined for 48-bit, 64-bit, and >64-bit instructions.
- Description of a 128-bit address space variant, **RV128**, has been added.
- Major opcodes in the 32-bit base instruction format have been allocated for user-defined custom extensions.
- A typographical error that suggested that stores source their data from *rd* has been corrected to refer to *rs2*.

Chapter 1. Introduction

RISC-V (pronounced "risk-five") is a new instruction-set architecture (ISA) that was originally designed to support computer architecture research and education, but which we now hope will also become a standard free and open architecture for industry implementations. Our goals in defining RISC-V include:

- A completely *open* ISA that is freely available to academia and industry.
- A *real* ISA suitable for direct native hardware implementation, not just simulation or binary translation.
- An ISA that avoids "over-architecting" for a particular microarchitecture style (e.g., microcoded, in-order, decoupled, out-of-order) or implementation technology (e.g., full-custom, ASIC, FPGA), but which allows efficient implementation in any of these.
- An ISA separated into a *small* base integer ISA, usable by itself as a base for customized accelerators or for educational purposes, and optional standard extensions, to support general-purpose software development.
- Support for the revised 2008 IEEE-754 floating-point standard. ([ANSI/IEEE Std 754-2008](#), [IEEE Standard for Floating-Point Arithmetic, 2008](#))
- An ISA supporting extensive ISA extensions and specialized variants.
- Both 32-bit and 64-bit address space variants for applications, operating system kernels, and hardware implementations.
- An ISA with support for highly parallel multicore or manycore implementations, including heterogeneous multiprocessors.
- Optional *variable-length instructions* to both expand available instruction encoding space and to support an optional *dense instruction encoding* for improved performance, static code size, and energy efficiency.
- A fully virtualizable ISA to ease hypervisor development.
- An ISA that simplifies experiments with new privileged architecture designs.



Commentary on our design decisions is formatted as in this paragraph. This non-normative text can be skipped if the reader is only interested in the specification itself.



The name RISC-V was chosen to represent the fifth major RISC ISA design from UC Berkeley (RISC-I ([Patterson & Séquin, 1981](#)), RISC-II ([Katevenis et al., 1983](#)), SOAR ([Ungar et al., 1984](#)), and SPUR ([Lee et al., 1989](#)) were the first four). We also pun on the use of the Roman numeral "V" to signify "variations" and "vectors", as support for a range of architecture research, including various data-parallel accelerators, is an explicit goal of the ISA design.

The RISC-V ISA is defined avoiding implementation details as much as possible (although commentary is included on implementation-driven decisions) and should be read as the software-visible interface to a wide variety of implementations rather than as the design of a particular hardware artifact. The RISC-V manual is structured in two volumes. This volume covers the design of the base *unprivileged* instructions, including optional unprivileged ISA extensions. Unprivileged instructions are those that are generally usable in all privilege modes in all privileged architectures, though behavior might vary depending on privilege mode and privilege architecture. The second volume provides the design of the first ("classic") privileged architecture. The manuals use IEC 80000-13:2008 conventions, with a byte of 8 bits.



In the unprivileged ISA design, we tried to remove any dependence on particular microarchitectural features, such as cache line size, or on privileged architecture details, such as page translation. This is both for simplicity and to allow maximum flexibility for alternative microarchitectures or alternative privileged architectures.

1.1. RISC-V Hardware Platform Terminology

A RISC-V hardware platform can contain one or more RISC-V-compatible processing cores together with other non-RISC-V-compatible cores, fixed-function accelerators, various physical memory structures, I/O devices, and an interconnect structure to allow the components to communicate.

A component is termed a *core* if it contains an independent instruction fetch unit. A RISC-V-compatible core might support multiple RISC-V-compatible hardware threads, or *harts*, through multithreading.

A RISC-V core might have additional specialized instruction-set extensions or an added *coprocessor*. We use the term *coprocessor* to refer to a unit that is attached to a RISC-V core and is mostly sequenced by a RISC-V instruction stream, but which contains additional architectural state and instruction-set extensions, and possibly some limited autonomy relative to the primary RISC-V instruction stream.

We use the term *accelerator* to refer to either a non-programmable fixed-function unit or a core that can operate autonomously but is specialized for certain tasks. In RISC-V systems, we expect many programmable accelerators will be RISC-V-based cores with specialized instruction-set extensions and/or customized coprocessors. An important class of RISC-V accelerators are I/O accelerators, which offload I/O processing tasks from the main application cores.

The system-level organization of a RISC-V hardware platform can range from a single-core microcontroller to a many-thousand-node cluster of shared-memory manycore server nodes. Even small systems-on-a-chip might be structured as a hierarchy of multicomputers and/or multiprocessors to modularize development effort or to provide secure isolation between subsystems.

1.2. RISC-V Software Execution Environments and Harts

The behavior of a RISC-V program depends on the execution environment in which it runs. A RISC-V execution environment interface (EEI) defines the initial state of the program, the number and type of harts in the environment including the privilege modes supported by the harts, the accessibility and attributes of memory and I/O regions, the behavior of all legal instructions executed on each hart (i.e., the ISA is one component of the EEI), and the handling of any interrupts or exceptions raised during execution including environment calls. Examples of EEIs include the Linux application binary interface (ABI), or the RISC-V supervisor binary interface (SBI). The implementation of a RISC-V execution environment can be pure hardware, pure software, or a combination of hardware and software. For example, opcode traps and software emulation can be used to implement functionality not provided in hardware. Examples of execution environment implementations include:

- "Bare metal" hardware platforms where harts are directly implemented by physical processor threads and instructions have full access to the physical address space. The hardware platform defines an execution environment that begins at power-on reset.
- RISC-V operating systems that provide multiple user-level execution environments by multiplexing user-level harts onto available physical processor threads and by controlling access to memory via virtual memory.
- RISC-V hypervisors that provide multiple supervisor-level execution environments for guest operating systems.
- RISC-V emulators, such as Spike, QEMU or rv8, which emulate RISC-V harts on an underlying x86 system, and which can provide either a user-level or a supervisor-level execution environment.



A bare hardware platform can be considered to define an EEI, where the accessible harts, memory, and other devices populate the environment, and the initial state is that at power-on reset. Generally, most software is designed to use a more abstract interface to the hardware, as more abstract EEIs provide greater portability across different hardware platforms. Often EEIs

are layered on top of one another, where one higher-level EEI uses another lower-level EEI.

From the perspective of software running in a given execution environment, a hart is a resource that autonomously fetches and executes RISC-V instructions within that execution environment. In this respect, a hart behaves like a hardware thread resource even if time-multiplexed onto real hardware by the execution environment. Some EEIs support the creation and destruction of additional harts, for example, via environment calls to fork new harts.

The execution environment is responsible for ensuring the eventual forward progress of each of its harts. For a given hart, that responsibility is suspended while the hart is exercising a mechanism that explicitly waits for an event, such as the wait-for-interrupt instruction defined in Volume II of this specification; and that responsibility ends if the hart is terminated. The following events constitute forward progress:

- The retirement of an instruction.
- A trap, as defined in [Section 1.6](#).
- Any other event defined by an extension to constitute forward progress.

The term hart was introduced in the work on Lithe ([Pan et al., 2009](#)) and ([Pan et al., 2010](#)) to provide a term to represent an abstract execution resource as opposed to a software thread programming abstraction.



The important distinction between a hardware thread (hart) and a software thread context is that the software running inside an execution environment is not responsible for causing progress of each of its harts; that is the responsibility of the outer execution environment. So the environment's harts operate like hardware threads from the perspective of the software inside the execution environment.

An execution environment implementation might time-multiplex a set of guest harts onto fewer host harts provided by its own execution environment but must do so in a way that guest harts operate like independent hardware threads. In particular, if there are more guest harts than host harts then the execution environment must be able to preempt the guest harts and must not wait indefinitely for guest software on a guest hart to "yield" control of the guest hart.

1.3. RISC-V ISA Overview

A RISC-V ISA is defined as a base integer ISA, which must be present in any implementation, plus optional extensions to the base ISA. The base integer ISAs are very similar to that of the early RISC processors except with no branch delay slots and with support for optional variable-length instruction encodings. A base is carefully restricted to a minimal set of instructions sufficient to provide a reasonable target for compilers, assemblers, linkers, and operating systems (with additional privileged operations), and so provides a convenient ISA and software toolchain "skeleton" around which more customized processor ISAs can be built.

Although it is convenient to speak of *the* RISC-V ISA, RISC-V is actually a family of related ISAs, of which there are currently four base ISAs. Each base integer instruction set is characterized by the width of the integer registers and the corresponding size of the address space and by the number of integer registers. There are two primary base integer variants, RV32I and RV64I, described in [Chapter 2](#) and [Chapter 4](#), which provide 32-bit or 64-bit address spaces respectively. We use the term XLEN to refer to the width of an integer register in bits (either 32 or 64). [Chapter 3](#) describes the RV32E and RV64E subset variants of the RV32I or RV64I base instruction sets respectively, which have been added to support small microcontrollers, and which have half the number of integer registers. The base integer instruction sets use a two's-complement representation for signed integer values.



Although 64-bit address spaces are a requirement for larger systems, we believe 32-bit address spaces will remain adequate for many embedded and client devices for decades to come and will be desirable to lower memory traffic and energy consumption. In addition, 32-bit address spaces are sufficient for educational purposes. A larger flat 128-bit address space might eventually be required and could be accommodated with a new RV128I base ISA within the existing RISC-V ISA framework.

The four base ISAs in RISC-V are treated as distinct base ISAs. A common question is why is there not a single ISA, and in particular, why is RV32I not a strict subset of RV64I? Some earlier ISA designs (SPARC, MIPS) adopted a strict superset policy when increasing address space size to support running existing 32-bit binaries on new 64-bit hardware.

The main advantage of explicitly separating base ISAs is that each base ISA can be optimized for its needs without requiring to support all the operations needed for other base ISAs. For example, RV64I can omit instructions and CSRs that are only needed to cope with the narrower registers in RV32I. The RV32I variants can use encoding space otherwise reserved for instructions only required by wider address-space variants.

The main disadvantage of not treating the design as a single ISA is that it complicates the hardware needed to emulate one base ISA on another (e.g., RV32I on RV64I). However, differences in addressing and illegal-instruction traps generally mean some mode switch would be required in hardware in any case even with full superset instruction encodings, and the different RISC-V base ISAs are similar enough that supporting multiple versions is relatively low cost. Although some have proposed that the strict superset design would allow legacy 32-bit libraries to be linked with 64-bit code, this is impractical in practice, even with compatible encodings, due to the differences in software calling conventions and system-call interfaces.



The RISC-V privileged architecture provides fields in `misa` to control the unprivileged ISA at each level to support emulating different base ISAs on the same hardware. We note that newer SPARC and MIPS ISA revisions have deprecated support for running 32-bit code unchanged on 64-bit systems.

A related question is why there is a different encoding for 32-bit adds in RV32I (ADD) and RV64I (ADDW)? The ADDW opcode could be used for 32-bit adds in RV32I and ADDD for 64-bit adds in RV64I, instead of the existing design which uses the same opcode ADD for 32-bit adds in RV32I and 64-bit adds in RV64I with a different opcode ADDW for 32-bit adds in RV64I. This would also be more consistent with the use of the same LW opcode for 32-bit load in both RV32I and RV64I. The very first versions of RISC-V ISA did have a variant of this alternate design, but the RISC-V design was changed to the current choice in January 2011. Our focus was on supporting 32-bit integers in the 64-bit ISA, not on providing compatibility with the 32-bit ISA, and the motivation was to remove the asymmetry that arose from having not all opcodes in RV32I have a `*W` suffix (e.g., ADDW, but AND not ANDW). In hindsight, this was perhaps not well-justified and a consequence of designing both ISAs at the same time as opposed to adding one later to sit on top of another, and also from a belief we had to fold platform requirements into the ISA spec which would imply that all the RV32I instructions would have been required in RV64I. It is too late to change the encoding now, but this is also of little practical consequence for the reasons stated above.

It has been noted we could enable the `*W` variants as an extension to RV32I systems to provide a common encoding across RV64I and a future RV32 variant.

RISC-V has been designed to support extensive customization and specialization. Each base integer ISA can be extended with one or more optional instruction-set extensions. An extension may be categorized as either standard, custom, or non-conforming. For this purpose, we divide each RISC-V instruction-set

encoding space (and related encoding spaces such as the CSRs) into three disjoint categories: *standard*, *reserved*, and *custom*. Standard extensions and encodings are defined by RISC-V International; any extensions not defined by RISC-V International are *non-standard*. Each base ISA and its standard extensions use only standard encodings, and shall not conflict with each other in their uses of these encodings. Reserved encodings are currently not defined but are saved for future standard extensions; once thus used, they become standard encodings. Custom encodings shall never be used for standard extensions and are made available for vendor-specific non-standard extensions. Non-standard extensions are either custom extensions, that use only custom encodings, or *non-conforming* extensions, that use any standard or reserved encoding. Instruction-set extensions are generally shared but may provide slightly different functionality depending on the base ISA. We have also developed a naming convention for RISC-V base instructions and instruction-set extensions, described in detail in [Chapter 37](#).

To support more general software development, a set of standard extensions are defined to provide integer multiply/divide, atomic operations, and single and double-precision floating-point arithmetic. The base integer ISA is named "I" (prefixed by RV32 or RV64 depending on integer register width), and contains integer computational instructions, integer loads, integer stores, and control-flow instructions. The standard integer multiplication and division extension is named "M", and adds instructions to multiply and divide values held in the integer registers. The standard atomic instruction extension, denoted by "A", adds instructions that atomically read, modify, and write memory for inter-processor synchronization. The standard single-precision floating-point extension, denoted by "F", adds floating-point registers, single-precision computational instructions, and single-precision loads and stores. The standard double-precision floating-point extension, denoted by "D", expands the floating-point registers, and adds double-precision computational instructions, loads, and stores. The standard "C" compressed instruction extension provides narrower 16-bit forms of common instructions.

Beyond the base integer ISA and these standard extensions, we believe it is rare that a new instruction will provide a significant benefit for all applications, although it may be very beneficial for a certain domain. As energy efficiency concerns are forcing greater specialization, we believe it is important to simplify the required portion of an ISA specification. Whereas other architectures usually treat their ISA as a single entity, which changes to a new version as instructions are added over time, RISC-V will endeavor to keep the base and each standard extension constant over time, and instead layer new instructions as further optional extensions. For example, the base integer ISAs will continue as fully supported standalone ISAs, regardless of any subsequent extensions.

1.4. Memory

A RISC-V hart has a single byte-addressable address space of 2^{XLEN} bytes for all memory accesses. A *word* of memory is defined as 32 bits (4 bytes). Correspondingly, a *halfword* is 16 bits (2 bytes), a *doubleword* is 64 bits (8 bytes), and a *quadword* is 128 bits (16 bytes). The memory address space is circular, so that the byte at address $2^{\text{XLEN}}-1$ is adjacent to the byte at address zero. Accordingly, memory address computations done by the hardware ignore overflow and instead wrap around modulo 2^{XLEN} .

The execution environment determines the mapping of hardware resources into a hart's address space. Different address ranges of a hart's address space may (1) contain *main memory*, or (2) contain one or more *I/O devices*. Reads and writes of I/O devices may have visible side effects, but accesses to main memory cannot. Vacant address ranges are not a separate category but can be represented as either main memory or I/O regions that are not accessible. Although it is possible for the execution environment to call everything in a hart's address space an I/O device, it is usually expected that some portion will be specified as main memory.

When a RISC-V platform has multiple harts, the address spaces of any two harts may be entirely the same, or entirely different, or may be partly different but sharing some subset of resources, mapped into the same or different address ranges.



For a purely "bare metal" environment, all harts may see an identical address space, accessed entirely by physical addresses. However, when the execution environment includes an operating system employing address translation, it is common for each hart to be given a virtual address space that is largely or entirely its own.

Executing each RISC-V machine instruction entails one or more memory accesses, subdivided into *implicit* and *explicit* accesses. For each instruction executed, an *implicit* memory read (instruction fetch) is done to obtain the encoded instruction to execute. Many RISC-V instructions perform no further memory accesses beyond instruction fetch. Specific load and store instructions perform an *explicit* read or write of memory at an address determined by the instruction. The execution environment may dictate that instruction execution performs other *implicit* memory accesses (such as to implement address translation) beyond those documented for the unprivileged ISA.

The execution environment determines what portions of the address space are accessible for each kind of memory access. For example, the set of locations that can be implicitly read for instruction fetch may or may not have any overlap with the set of locations that can be explicitly read by a load instruction; and the set of locations that can be explicitly written by a store instruction may be only a subset of locations that can be read. Ordinarily, if an instruction attempts to access memory at an inaccessible address, an exception is raised for the instruction.

Except when specified otherwise, implicit reads that do not raise an exception and that have no side effects may occur arbitrarily early and speculatively, even before the machine could possibly prove that the read will be needed. For instance, a valid implementation could attempt to read all of main memory at the earliest opportunity, cache as many fetchable (executable) bytes as possible for later instruction fetches, and avoid reading main memory for instruction fetches ever again. To ensure that certain implicit reads are ordered only after writes to the same memory locations, software must execute specific fence or cache-control instructions defined for this purpose (such as the FENCE.I instruction defined in [Chapter 5](#)).

The memory accesses (implicit or explicit) made by a hart may appear to occur in a different order as perceived by another hart or by any other agent that can access the same memory. This perceived reordering of memory accesses is always constrained, however, by the applicable memory consistency model. The default memory consistency model for RISC-V is the RISC-V Weak Memory Ordering (RVWMO), defined in [Chapter 18](#) and in appendices. Optionally, an implementation may adopt the stronger model of Total Store Ordering, as defined in [Chapter 19](#). The execution environment may also add constraints that further limit the perceived reordering of memory accesses. Since the RVWMO model is the weakest model allowed for any RISC-V implementation, software written for this model is compatible with the actual memory consistency rules of all RISC-V implementations. As with implicit reads, software must execute fence or cache-control instructions to ensure specific ordering of memory accesses beyond the requirements of the assumed memory consistency model and execution environment.

1.5. Base Instruction-Length Encoding

The base RISC-V ISA has fixed-length 32-bit instructions that must be naturally aligned on 32-bit boundaries. However, the standard RISC-V encoding scheme is designed to support ISA extensions with variable-length instructions, where each instruction can be any number of 16-bit instruction *parcels* in length and parcels are naturally aligned on 16-bit boundaries. The standard compressed ISA extension described in [Chapter 28](#) reduces code size by providing compressed 16-bit instructions and relaxes the alignment constraints to allow all instructions (16 bit and 32 bit) to be aligned on any 16-bit boundary to improve code density.

We use the term IALIGN (measured in bits) to refer to the instruction-address alignment constraint the implementation enforces. IALIGN is 32 bits in the base ISA, but some ISA extensions, including the

compressed ISA extension, relax IALIGN to 16 bits. IALIGN may not take on any value other than 16 or 32.

We use the term ILEN (measured in bits) to refer to the maximum instruction length supported by an implementation, and which is always a multiple of IALIGN. For implementations supporting only a base instruction set, ILEN is 32 bits. Implementations supporting longer instructions have larger values of ILEN.

All the 32-bit instructions in the base ISA have their lowest two bits set to **11**. The optional compressed 16-bit instruction-set extensions have their lowest two bits equal to **00**, **01**, or **10**.



Given the code size and energy savings of a compressed format, we wanted to build in support for a compressed format to the ISA encoding scheme rather than adding this as an afterthought, but to allow simpler implementations we didn't want to make the compressed format mandatory. We also wanted to optionally allow longer instructions to support experimentation and larger instruction-set extensions. Although our encoding convention required a tighter encoding of the core RISC-V ISA, this has several beneficial effects.

An implementation of the standard IMAFD ISA need only hold the most-significant 30 bits in instruction caches (a 6.25% saving). On instruction cache refills, any instructions encountered with either low bit clear should be recoded into illegal 30-bit instructions before storing in the cache to preserve illegal-instruction exception behavior.

Perhaps more importantly, by condensing our base ISA into a subset of the 32-bit instruction word, we leave more space available for non-standard and custom extensions. In particular, the base RV32I ISA uses less than 1/8 of the encoding space in the 32-bit instruction word. An implementation that does not require support for the standard compressed instruction extension can map 3 additional non-conforming 30-bit instruction spaces into the 32-bit fixed-width format, while preserving support for standard ≥ 32 -bit instruction-set extensions.

Encodings with bits [15:0] all zeros are defined as illegal instructions. These instructions are considered to be of minimal length: 16 bits if any 16-bit instruction-set extension is present, otherwise 32 bits. The encoding with bits [ILEN-1:0] all ones is also illegal; this instruction is considered to be ILEN bits long.



We consider it a feature that any length of instruction containing all zero bits is not legal, as this quickly traps erroneous jumps into zeroed memory regions. Similarly, we also reserve the instruction encoding containing all ones to be an illegal instruction, to catch the other common pattern observed with unprogrammed non-volatile memory devices, disconnected memory buses, or broken memory devices.

Software can rely on a naturally aligned 32-bit word containing zero to act as an illegal instruction on all RISC-V implementations, to be used by software where an illegal instruction is explicitly desired. Defining a corresponding known illegal value for all ones is more difficult due to the variable-length encoding. Software cannot generally use the illegal value of ILEN bits of all 1s, as software might not know ILEN for the eventual target machine (e.g., if software is compiled into a standard binary library used by many different machines). Defining a 32-bit word of all ones as illegal was also considered, as all machines must support a 32-bit instruction size, but this requires the instruction-fetch unit on machines with ILEN > 32 report an illegal-instruction exception rather than an access-fault exception when such an instruction borders a protection boundary, complicating variable-instruction-length fetch and decode.

RISC-V base ISAs have either little-endian or big-endian memory systems, with the privileged architecture further defining bi-endian operation. Instructions are stored in memory as a sequence of 16-bit little-endian parcels, regardless of memory system endianness. Parcels forming one instruction are stored at increasing halfword addresses, with the lowest-addressed parcel holding the lowest-numbered bits in the instruction specification.

We originally chose little-endian byte ordering for the RISC-V memory system because little-endian systems are currently dominant commercially (all x86 systems; iOS, Android, and Windows for ARM). A minor point is that we have also found little-endian memory systems to be more natural for hardware designers. However, certain application areas, such as IP networking, operate on big-endian data structures, and certain legacy code bases have been built assuming big-endian processors, so we have defined big-endian and bi-endian variants of RISC-V.

We have to fix the order in which instruction parcels are stored in memory, independent of memory system endianness, to ensure that the length-encoding bits always appear first in halfword address order. This allows the length of a variable-length instruction to be quickly determined by an instruction-fetch unit by examining only the first few bits of the first 16-bit instruction parcel.



We further make the instruction parcels themselves little-endian to decouple the instruction encoding from the memory system endianness altogether. This design benefits both software tooling and bi-endian hardware. Otherwise, for instance, a RISC-V assembler or disassembler would always need to know the intended active endianness, despite that in bi-endian systems, the endianness mode might change dynamically during execution. In contrast, by giving instructions a fixed endianness, it is sometimes possible for carefully written software to be endianness-agnostic even in binary form, much like position-independent code.

The choice to have instructions be only little-endian does have consequences, however, for RISC-V software that encodes or decodes machine instructions. Big-endian JIT compilers, for example, must swap the byte order when storing to instruction memory.

Once we had decided to fix on a little-endian instruction encoding, this naturally led to placing the length-encoding bits in the LSB positions of the instruction format to avoid breaking up opcode fields.

1.6. Exceptions, Traps, and Interrupts

We use the term *exception* to refer to an unusual condition occurring at run time associated with an instruction in the current RISC-V hart. We use the term *interrupt* to refer to an external asynchronous event that may cause a RISC-V hart to experience an unexpected transfer of control. We use the term *trap* to refer to the transfer of control to a trap handler caused by either an exception or an interrupt.

The instruction descriptions in following chapters describe conditions that can raise an exception during execution. The general behavior of most RISC-V EEIs is that a trap to some handler occurs when an exception is signaled on an instruction (except for floating-point exceptions, which, in the standard floating-point extensions, do not cause traps). The manner in which interrupts are generated, routed to, and enabled by a hart depends on the EEI.



Our use of "exception" and "trap" is compatible with that in the IEEE-754 floating-point standard.

How traps are handled and made visible to software running on the hart depends on the enclosing execution environment. From the perspective of software running inside an execution environment, traps encountered by a hart at runtime can have four different effects:

Contained Trap

The trap is visible to, and handled by, software running inside the execution environment. For example, in an EEI providing both supervisor and user mode on harts, an ECALL by a user-mode hart will generally result in a transfer of control to a supervisor-mode handler running on the same hart.

Similarly, in the same environment, when a hart is interrupted, an interrupt handler will be run in supervisor mode on the hart.

Requested Trap

The trap is a synchronous exception that is an explicit call to the execution environment requesting an action on behalf of software inside the execution environment. An example is a system call. In this case, execution may or may not resume on the hart after the requested action is taken by the execution environment. For example, a system call could remove the hart or cause an orderly termination of the entire execution environment.

Invisible Trap

The trap is handled transparently by the execution environment and execution resumes normally after the trap is handled. Examples include emulating missing instructions, handling non-resident page faults in a demand-paged virtual-memory system, or handling device interrupts for a different job in a multiprogrammed machine. In these cases, the software running inside the execution environment is not aware of the trap (we ignore timing effects in these definitions).

Fatal Trap

The trap represents a fatal failure and causes the execution environment to terminate execution. Examples include failing a virtual-memory page-protection check or allowing a watchdog timer to expire. Each EEI should define how execution is terminated and reported to an external environment.

Table 1 shows the characteristics of each kind of trap.

Table 1. Characteristics of traps

	Contained	Requested	Invisible	Fatal
Execution terminates	No	No ¹	No	Yes
Software is oblivious	No	No	Yes	Yes ²
Handled by environment	No	Yes	Yes	Yes

¹ Termination may be requested

² Imprecise fatal traps might be observable by software

The EEI defines for each trap whether it is handled precisely, though the recommendation is to maintain preciseness where possible. Contained and requested traps can be observed to be imprecise by software inside the execution environment. Invisible traps, by definition, cannot be observed to be precise or imprecise by software running inside the execution environment. Fatal traps can be observed to be imprecise by software running inside the execution environment, if known-errorful instructions do not cause immediate termination.

Because this document describes unprivileged instructions, traps are rarely mentioned. Architectural means to handle contained traps are defined in the privileged architecture manual, along with other features to support richer EEIs. Unprivileged instructions that are defined solely to cause requested traps are documented here. Invisible traps are, by their nature, out of scope for this document. Instruction encodings that are not defined here and not defined by some other means may cause a fatal trap.

1.7. UNSPECIFIED Behaviors and Values

The architecture fully describes what implementations must do and any constraints on what they may do. In cases where the architecture intentionally does not constrain implementations, the term UNSPECIFIED is explicitly used.

The term UNSPECIFIED refers to a behavior or value that is intentionally unconstrained. The definition of these behaviors or values is open to extensions, platform standards, or implementations. Extensions, platform standards, or implementation documentation may provide normative content to further constrain cases that the base architecture defines as UNSPECIFIED.

Like the base architecture, extensions should fully describe allowable behavior and values and use the term UNSPECIFIED for cases that are intentionally unconstrained. These cases may be constrained or defined by other extensions, platform standards, or implementations.

Chapter 2. RV32I Base Integer Instruction Set, Version 2.1

This chapter describes the RV32I base integer instruction set.



RV32I was designed to be sufficient to form a compiler target and to support modern operating system environments. The ISA was also designed to reduce the hardware required in a minimal implementation. RV32I contains 40 unique instructions, though a simple implementation might cover the ECALL/EBREAK instructions with a single SYSTEM hardware instruction that always traps and might be able to implement the FENCE instruction as a NOP, reducing base instruction count to 38 total. RV32I can emulate almost any other ISA extension (except the A extension, which requires additional hardware support for atomicity).

In practice, a hardware implementation including the machine-mode privileged architecture will also require the 6 CSR instructions.

Subsets of the base integer ISA might be useful for pedagogical purposes, but the base has been defined such that there should be little incentive to subset a real hardware implementation beyond omitting support for misaligned memory accesses and treating all SYSTEM instructions as a single trap.



The standard RISC-V assembly language syntax is documented in the Assembly Programmer's Manual ([RISC-V Assembly Programmer's Manual](#), n.d.).



Most of the commentary for RV32I also applies to the RV64I base.

2.1. Programmers' Model for Base Integer ISA

[Table 2](#) shows the unprivileged state for the base integer ISA. For RV32I, the 32 **x** registers are each 32 bits wide, i.e., **XLEN=32**. Register **x0** is hardwired with all bits equal to 0. General purpose registers **x1-x31** hold values that various instructions interpret as a collection of Boolean values, or as two's complement signed binary integers or unsigned binary integers.

There is one additional unprivileged register: the program counter **pc** holds the address of the current instruction.

Table 2. RISC-V base unprivileged integer register state.

XLEN-1	0
x0/zero	
x1	
x2	
x3	
x4	
x5	
x6	
x7	
x8	
x9	
x10	
x11	
x12	
x13	

XLEN-1	0
x14	
x15	
x16	
x17	
x18	
x19	
x20	
x21	
x22	
x23	
x24	
x25	
x26	
x27	
x28	
x29	
x30	
x31	
pc	

There is no dedicated stack pointer or subroutine return address link register in the Base Integer ISA; the instruction encoding allows any **x** register to be used for these purposes. However, the standard software calling convention uses register **x1** to hold the return address for a call, with register **x5** available as an alternate link register. The standard calling convention uses register **x2** as the stack pointer.

Hardware might choose to accelerate function calls and returns that use **x1** or **x5**. See the descriptions of the JAL and JALR instructions.

The optional compressed 16-bit instruction format is designed around the assumption that **x1** is the return address register and **x2** is the stack pointer. Software using other conventions will operate correctly but may have greater code size.



The number of available architectural registers can have large impacts on code size, performance, and energy consumption. Although 16 registers would arguably be sufficient for an integer ISA running compiled code, it is impossible to encode a complete ISA with 16 registers in 16-bit instructions using a 3-address format. Although a 2-address format would be possible, it would increase instruction count and lower efficiency. We wanted to avoid intermediate instruction sizes (such as Xtensa's 24-bit instructions) to simplify base hardware implementations, and once a 32-bit instruction size was adopted, it was straightforward to support 32 integer registers. A larger number of integer registers also helps performance on high-performance code, where there can be extensive use of loop unrolling, software pipelining, and cache tiling.

For these reasons, we chose a conventional size of 32 integer registers for RV32I. Dynamic register usage tends to be dominated by a few frequently accessed registers, and register file implementations can be optimized to reduce access energy for the frequently accessed registers (Tseng & Asanović, 2000). The optional compressed 16-bit instruction format mostly only accesses 8 registers and hence can provide a dense instruction encoding, while additional instruction-set extensions could support a much larger register space (either flat or

hierarchical) if desired.

For resource-constrained embedded applications, we have defined the RV32E subset, which only has 16 registers ([Chapter 3](#)).

2.2. Base Instruction Formats

In the base RV32I ISA, there are four core instruction formats (R/I/S/U), as shown in [Base instruction formats](#). All are a fixed 32 bits in length. The base ISA has **IALIGN=32**, meaning that instructions must be aligned on a four-byte boundary in memory. An instruction-address-misaligned exception is generated on a taken branch or unconditional jump if the target address is not **IALIGN-bit** aligned. This exception is reported on the branch or jump instruction, not on the target instruction. No instruction-address-misaligned exception is generated for a conditional branch that is not taken.



The alignment constraint for base ISA instructions is relaxed to a two-byte boundary when instruction extensions with 16-bit lengths or other odd multiples of 16-bit lengths are added (i.e., **IALIGN=16**).

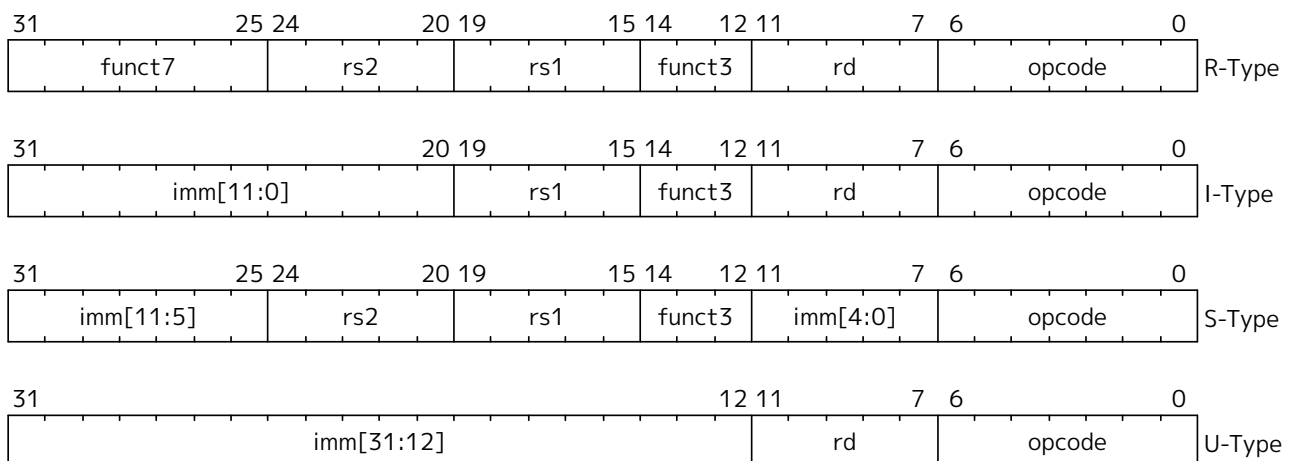
Instruction-address-misaligned exceptions are reported on the branch or jump that would cause instruction misalignment to help debugging, and to simplify hardware design for systems with **IALIGN=32**, where these are the only places where misalignment can occur.

The behavior upon decoding a reserved instruction is UNSPECIFIED.



Some platforms may require that opcodes reserved for standard use raise an illegal-instruction exception. Other platforms may permit reserved opcode space be used for non-conforming extensions.

The RISC-V ISA keeps the source (*rs1* and *rs2*) and destination (*rd*) registers at the same position in all formats to simplify decoding. Except for the 5-bit immediates used in CSR instructions ([Chapter 6](#)), immediates are always sign-extended, and are generally packed towards the leftmost available bits in the instruction and have been allocated to reduce hardware complexity. In particular, the sign bit for all immediates is always in bit 31 of the instruction to speed sign-extension circuitry.



RISC-V base instruction formats. Each immediate subfield is labeled with the bit position (*imm[x]*) in the immediate value being produced, rather than the bit position within the instruction's immediate field as is usually done.



Decoding register specifiers is usually on the critical paths in implementations, and so the instruction format was chosen to keep all register specifiers at the same position in all formats at the expense of having to move immediate bits across formats (a property shared with RISC-

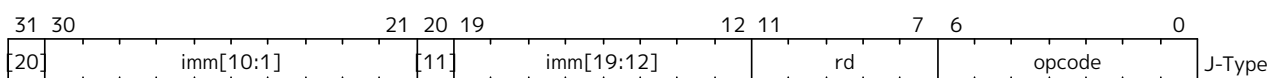
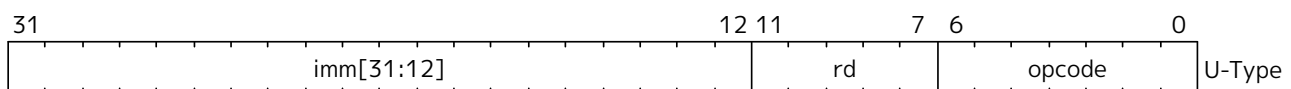
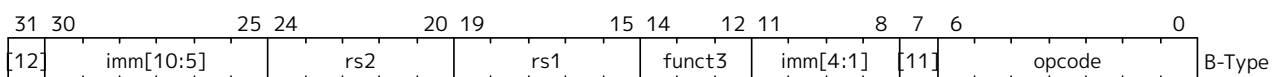
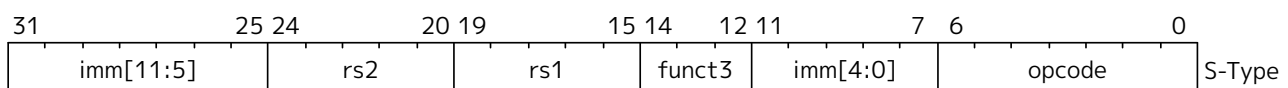
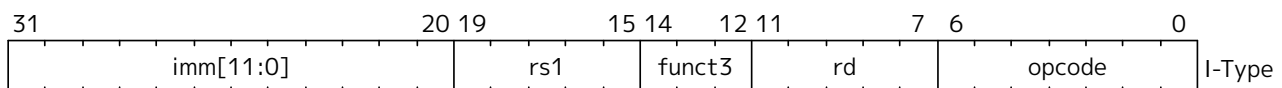
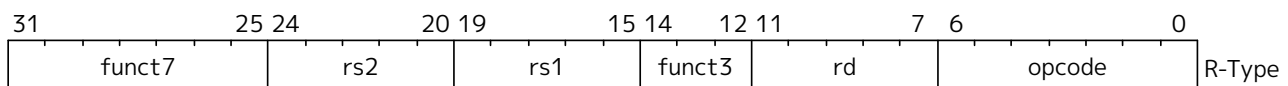
IV aka. SPUR ([Lee et al., 1989](#))).

In practice, most immediates are either small or require all XLEN bits. We chose an asymmetric immediate split (12 bits in regular instructions plus a special load-upper-immediate instruction with 20 bits) to increase the opcode space available for regular instructions.

Immediates are sign-extended because we did not observe a benefit to using zero extension for some immediates as in the MIPS ISA and wanted to keep the ISA as simple as possible.

2.3. Immediate Encoding Variants

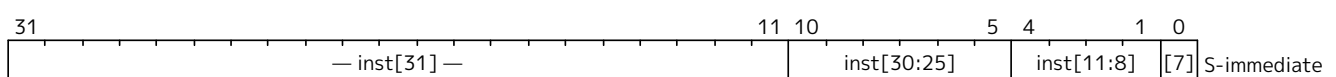
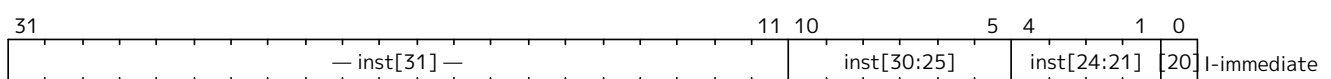
There are a further two variants of the instruction formats (B/J) based on the handling of immediates, as shown in [Base instruction formats immediate variants](#).



The only difference between the S and B formats is that the 12-bit immediate field is used to encode branch offsets in multiples of 2 in the B format. Instead of shifting all bits in the instruction-encoded immediate left by one in hardware as is conventionally done, the middle bits (imm[10:1]) and sign bit stay in fixed positions, while the lowest bit in S format (inst[7]) encodes a high-order bit in B format.

Similarly, the only difference between the U and J formats is that the 20-bit immediate is shifted left by 12 bits to form U immediates and by 1 bit to form J immediates. The location of instruction bits in the U and J format immediates is chosen to maximize overlap with the other formats and with each other.

[Immediate types](#) shows the immediates produced by each of the base instruction formats, and is labeled to show which instruction bit (inst[y]) produces each bit of the immediate value.



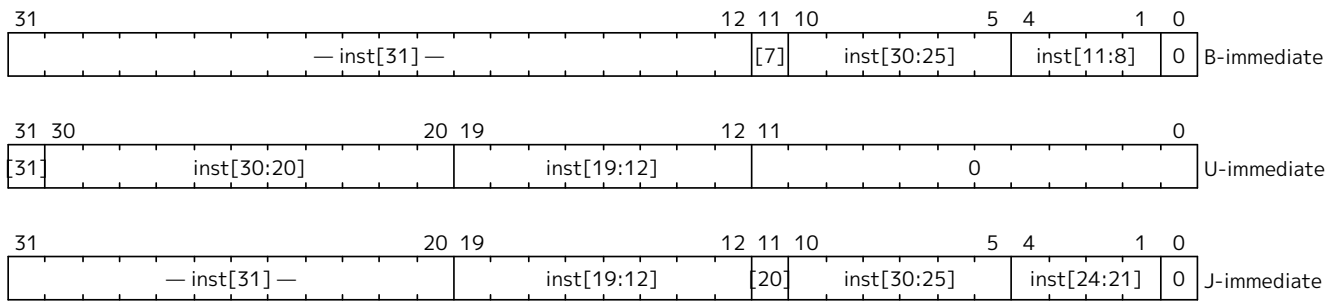


Figure 1. Types of immediate produced by RISC-V instructions.

The fields are labeled with the instruction bits used to construct their value. Sign extensions always uses `inst[31]`.



Sign extension is one of the most critical operations on immediates (particularly for $XLEN > 32$), and in RISC-V the sign bit for all immediates is always held in bit 31 of the instruction to allow sign extension to proceed in parallel with instruction decoding.

Although more complex implementations might have separate adders for branch and jump calculations and so would not benefit from keeping the location of immediate bits constant across types of instruction, we wanted to reduce the hardware cost of the simplest implementations. By rotating bits in the instruction encoding of B and J immediates instead of using dynamic hardware multiplexers to multiply the immediate by 2, we reduce instruction signal fanout and immediate multiplexer costs by around a factor of 2. The scrambled immediate encoding will add negligible time to static or ahead-of-time compilation. For dynamic generation of instructions, there is some small additional overhead, but the most common short forward branches have straightforward immediate encodings.

2.4. Integer Computational Instructions

Most integer computational instructions operate on $XLEN$ bits of values held in the integer register file. Integer computational instructions are either encoded as register-immediate operations using the I-type format or as register-register operations using the R-type format. The destination is register `rd` for both register-immediate and register-register instructions. No integer computational instructions cause arithmetic exceptions.



We did not include special instruction-set support for overflow checks on integer arithmetic operations in the base instruction set, as many overflow checks can be cheaply implemented using RISC-V branches. Overflow checking for unsigned addition requires only a single additional branch instruction after the addition: `add t0, t1, t2; bltu t0, t1, overflow`.

For signed addition, if one operand's sign is known, overflow checking requires only a single branch after the addition: `addi t0, t1, +imm; blt t0, t1, overflow`. This covers the common case of addition with an immediate operand.

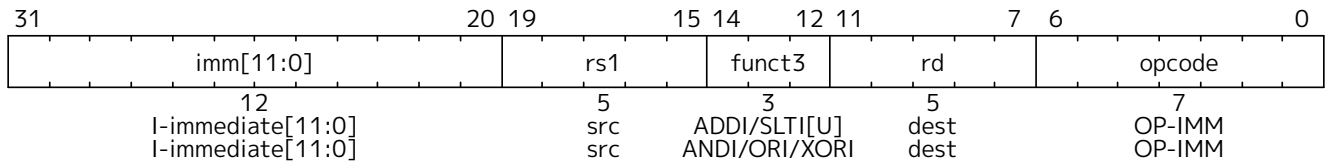
For general signed addition, three additional instructions after the addition are required, leveraging the observation that the sum should be less than one of the operands if and only if the other operand is negative.

```
add t0, t1, t2
slti t3, t2, 0
slt t4, t0, t1
```

```
bne t3, t4, overflow
```

In RV64I, checks of 32-bit signed additions can be optimized further by comparing the results of ADD and ADDW on the operands.

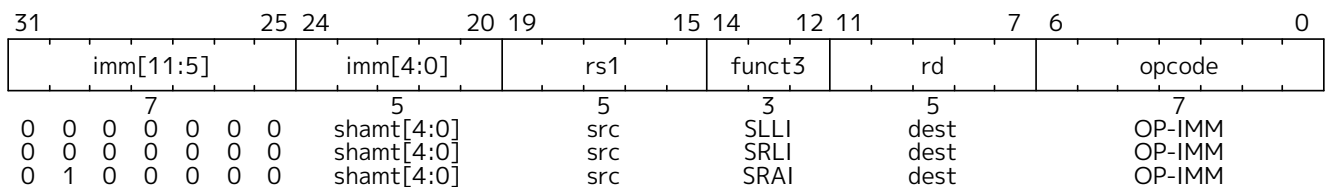
2.4.1. Integer Register-Immediate Instructions



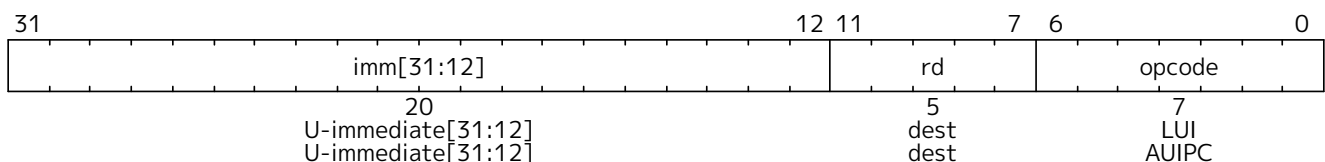
ADDI adds the sign-extended 12-bit immediate to register *rs1*. Arithmetic overflow is ignored and the result is simply the low XLEN bits of the result. ADDI *rd*, *rs1*, *O* is used to implement the MV *rd*, *rs1* assembler pseudoinstruction.

SLTI (set less than immediate) places the value 1 in register *rd* if register *rs1* is less than the sign-extended immediate when both are treated as signed numbers, else 0 is written to *rd*. SLTIU is similar but compares the values as unsigned numbers (i.e., the immediate is first sign-extended to XLEN bits then treated as an unsigned number). Note, SLTIU *rd*, *rs1*, 1 sets *rd* to 1 if *rs1* equals zero, otherwise sets *rd* to 0 (assembler pseudoinstruction SEQZ *rd*, *rs*).

ANDI, ORI, XORI are logical operations that perform bitwise AND, OR, and XOR on register *rs1* and the sign-extended 12-bit immediate and place the result in *rd*. Note, XORI *rd*, *rs1*, -1 performs a bitwise logical inversion of register *rs1* (assembler pseudoinstruction NOT *rd*, *rs*).



Shifts by a constant are encoded as a specialization of the I-type format. The operand to be shifted is in *rs1*, and the shift amount is encoded in the lower 5 bits of the I-immediate field. The right-shift type is encoded in bit 30. SLLI is a logical left shift (zeros are shifted into the lower bits); SRLI is a logical right shift (zeros are shifted into the upper bits); and SRAI is an arithmetic right shift (the original sign bit is copied into the vacated upper bits).



LUI (load upper immediate) is used to build 32-bit constants and uses the U-type format. LUI places the 32-bit U-immediate value into the destination register *rd*, filling in the lowest 12 bits with zeros.

AUIPC (add upper immediate to **pc**) is used to build **pc**-relative addresses and uses the U-type format. AUIPC forms a 32-bit offset from the U-immediate, filling in the lowest 12 bits with zeros, adds this offset to the address of the AUIPC instruction, then places the result in register *rd*.



The assembly syntax for `lui` and `auipc` does not represent the lower 12 bits of the U-immediate, which are always zero.

The AUIPC instruction supports two-instruction sequences to access arbitrary offsets from the PC for both control-flow transfers and data accesses. The combination of an AUIPC and the 12-bit immediate in a JALR can transfer control to any 32-bit PC-relative address, while an AUIPC plus the 12-bit immediate offset in regular load or store instructions can access any 32-bit PC-relative data address.

The current PC can be obtained by setting the U-immediate to 0. Although a JAL +4 instruction could also be used to obtain the local PC (of the instruction following the JAL), it might cause pipeline breaks in simpler microarchitectures or pollute BTB structures in more complex microarchitectures.

2.4.2. Integer Register-Register Instructions

RV32I defines several arithmetic R-type operations. All operations read the *rs1* and *rs2* registers as source operands and write the result into register *rd*. The *funct7* and *funct3* fields select the type of operation.

31								25		24		20				19		15		14		12		11		7		6		0	
funct7								rs2				rs1				funct3				rd				opcode							
7								5				5				3				5				7							
0	0	0	0	0	0	0	0	src2				src1				ADD/SLT[U]				dest				OP							
0	0	0	0	0	0	0	0	src2				src1				AND/OR/XOR				dest				OP							
0	0	0	0	0	0	0	0	src2				src1				SLL/SRL				dest				OP							
0	1	0	0	0	0	0	0	src2				src1				SUB/SRA				dest				OP							

ADD performs the addition of *rs1* and *rs2*. SUB performs the subtraction of *rs2* from *rs1*. Overflows are ignored and the low XLEN bits of results are written to the destination *rd*. SLT and SLTU perform signed and unsigned compares respectively, writing 1 to *rd* if *rs1* < *rs2*, 0 otherwise. Note, SLTU *rd*, *x0*, *rs2* sets *rd* to 1 if *rs2* is not equal to zero, otherwise sets *rd* to zero (assembler pseudoinstruction SNEZ *rd*, *rs*). AND, OR, and XOR perform bitwise logical operations.

SLL, SRL, and SRA perform logical left, logical right, and arithmetic right shifts on the value in register *rs1* by the shift amount held in the lower 5 bits of register *rs2*.

2.4.3. NOP Instruction

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The NOP instruction does not change any architecturally visible state, except for advancing the **pc** and incrementing any applicable performance counters. NOP is encoded as ADDI *x0*, *x0*, 0.

NOPs can be used to align code segments to microarchitecturally significant address boundaries, or to leave space for inline code modifications. Although there are many possible ways to encode a NOP, we define a canonical NOP encoding to allow microarchitectural optimizations as well as for more readable disassembly output. The other NOP encodings are made available for [HINT Instructions](#).



ADDI was chosen for the NOP encoding as this is most likely to take fewest resources to execute across a range of systems (if not optimized away in decode). In particular, the instruction only reads one register. Also, an ADDI functional unit is more likely to be available in a superscalar design as adds are the most common operation. In particular, address-generation functional units can execute ADDI using the same hardware needed for base+offset address calculations, while register-register ADD or logical/shift operations

require additional hardware.

2.5. Control Transfer Instructions

RV32I provides two types of control transfer instructions: unconditional jumps and conditional branches. Control transfer instructions in RV32I do *not* have architecturally visible delay slots.

If an instruction access-fault or instruction page-fault exception occurs on the target of a jump or taken branch, the exception is reported on the target instruction, not on the jump or branch instruction.

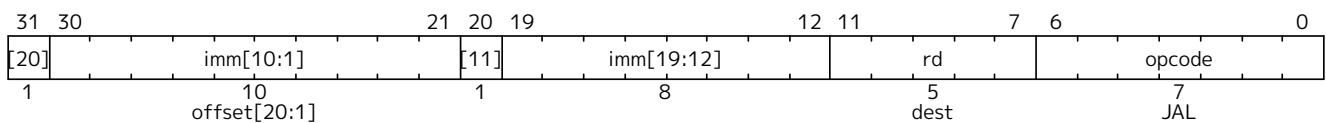
2.5.1. Unconditional Jumps

The jump and link (JAL) instruction uses the J-type format, where the J-immediate encodes a signed offset in multiples of 2 bytes. The offset is sign-extended and added to the address of the jump instruction to form the jump target address. Jumps can therefore target a ± 1 MiB range. JAL stores the address of the instruction following the jump ($pc+4$) into register *rd*. The standard software calling convention uses 'x1' as the return address register and 'x5' as an alternate link register.



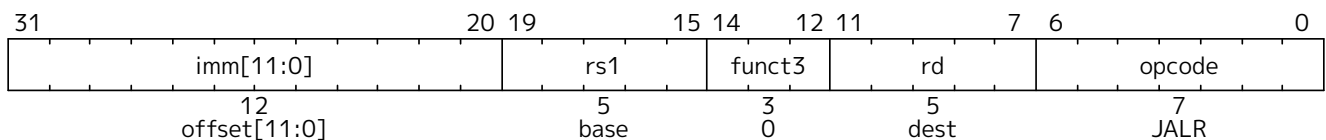
The alternate link register supports calling millicode routines (e.g., those to save and restore registers in compressed code) while preserving the regular return address register. The register x5 was chosen as the alternate link register as it maps to a temporary in the standard calling convention, and has an encoding that is only one bit different than the regular link register.

Plain unconditional jumps (assembler pseudoinstruction J) are encoded as a JAL with $rd=x0$.



The indirect jump instruction JALR (jump and link register) uses the I-type encoding. The target address is obtained by adding the sign-extended 12-bit I-immediate to the register *rs1*, then setting the least-significant bit of the result to zero. The address of the instruction following the jump ($pc+4$) is written to register *rd*. Register $x0$ can be used as the destination if the result is not required.

Plain unconditional indirect jumps (assembler pseudoinstruction JR) are encoded as a JALR with $rd=x0$. Procedure returns in the standard calling convention (assembler pseudoinstruction RET) are encoded as a JALR with $rd=x0$, $rs1=x1$, and $imm=0$.



*The unconditional jump instructions all use PC-relative addressing to help support position-independent code. The JALR instruction was defined to enable a two-instruction sequence to jump anywhere in a 32-bit absolute address range. A LUI instruction can first load *rs1* with the upper 20 bits of a target address, then JALR can add in the lower bits. Similarly, AUIPC then JALR can jump anywhere in a 32-bit pc-relative address range.*

Note that the JALR instruction does not treat the 12-bit immediate as multiples of 2 bytes, unlike the conditional branch instructions. This avoids one more immediate format in hardware. In practice, most uses of JALR will have either a zero immediate or be paired with a LUI or AUIPC, so the slight reduction in range is not significant.

Clearing the least-significant bit when calculating the JALR target address both simplifies the hardware slightly and allows the low bit of function pointers to be used to store auxiliary information. Although there is potentially a slight loss of error checking in this case, in practice jumps to an incorrect instruction address will usually quickly raise an exception.

When used with a base $rs1=x0$, JALR can be used to implement a single instruction subroutine call to the lowest or highest address region from anywhere in the address space, which could be used to implement fast calls to a small runtime library. Alternatively, an ABI could dedicate a general-purpose register to point to a library elsewhere in the address space.

The JAL and JALR instructions will generate an instruction-address-misaligned exception if the target address is not aligned to a four-byte boundary.



Instruction-address-misaligned exceptions are not possible on machines with $IALIGN=16$, e.g., those with the compressed instruction-set extension, C.

Return-address prediction stacks are a common feature of high-performance instruction-fetch units, but require accurate detection of instructions used for procedure calls and returns to be effective. For RISC-V, hints as to the instructions' usage are encoded implicitly via the register numbers used. A JAL instruction should push the return address onto a return-address stack (RAS) only when rd is 'x1' or x5. JALR instructions should push/pop a RAS as shown in Table 3.

Table 3. Return-address stack prediction hints encoded in the register operands of a JALR instruction.

rd is x1/x5	$rs1$ is x1/x5	$rd=rs1$	RAS action
No	No	—	None
No	Yes	—	Pop
Yes	No	—	Push
Yes	Yes	No	Pop, then push
Yes	Yes	Yes	Push

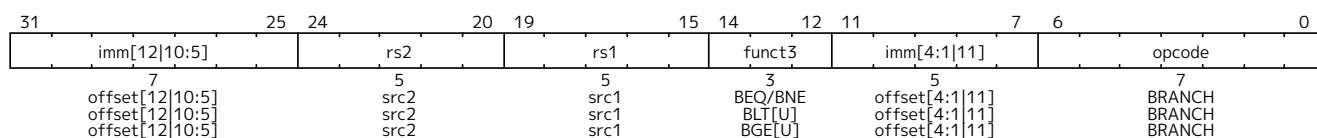
Some other ISAs added explicit hint bits to their indirect-jump instructions to guide return-address stack manipulation. We use implicit hinting tied to register numbers and the calling convention to reduce the encoding space used for these hints.



When two different link registers (x1 and x5) are given as $rs1$ and rd , then the RAS is both popped and pushed to support coroutines. If $rs1$ and rd are the same link register (either x1 or x5), the RAS is only pushed to enable macro-op fusion of the sequences: `lui ra, imm20; jalr ra, imm12(ra)` and `_auipc ra, imm20; jalr ra, imm12(ra)`

2.5.2. Conditional Branches

All branch instructions use the B-type instruction format. The 12-bit B-immediate encodes signed offsets in multiples of 2 bytes. The offset is sign-extended and added to the address of the branch instruction to give the target address. The conditional branch range is ± 4 KiB.



Branch instructions compare two registers. BEQ and BNE take the branch if registers $rs1$ and $rs2$ are equal or unequal respectively. BLT and BLTU take the branch if $rs1$ is less than $rs2$, using signed and unsigned comparison respectively. BGE and BGEU take the branch if $rs1$ is greater than or equal to $rs2$, using signed

and unsigned comparison respectively. Note, BGT, BGTU, BLE, and BLEU can be synthesized by reversing the operands to BLT, BLTU, BGE, and BGEU, respectively.



Signed array bounds may be checked with a single BLTU instruction, since any negative index will compare greater than any nonnegative bound.

Software should be optimized such that the sequential code path is the most common path, with less-frequently taken code paths placed out of line. Software should also assume that backward branches will be predicted taken and forward branches as not taken, at least the first time they are encountered. Dynamic predictors should quickly learn any predictable branch behavior.

Unlike some other architectures, the RISC-V jump (JAL with $rd=x0$) instruction should always be used for unconditional branches instead of a conditional branch instruction with an always-true condition. RISC-V jumps are also PC-relative and support a much wider offset range than branches, and will not pollute conditional-branch prediction tables.

The conditional branches were designed to include arithmetic comparison operations between two registers (as also done in PA-RISC, Xtensa, and MIPS R6), rather than use condition codes (x86, ARM, SPARC, PowerPC), or to only compare one register against zero (Alpha, MIPS), or two registers only for equality (MIPS). This design was motivated by the observation that a combined compare-and-branch instruction fits into a regular pipeline, avoids additional condition code state or use of a temporary register, and reduces static code size and dynamic instruction fetch traffic. Another point is that comparisons against zero require non-trivial circuit delay (especially after the move to static logic in advanced processes) and so are almost as expensive as arithmetic magnitude compares. Another advantage of a fused compare-and-branch instruction is that branches are observed earlier in the front-end instruction stream, and so can be predicted earlier. There is perhaps an advantage to a design with condition codes in the case where multiple branches can be taken based on the same condition codes, but we believe this case to be relatively rare.

We considered but did not include static branch hints in the instruction encoding. These can reduce the pressure on dynamic predictors, but require more instruction encoding space and software profiling for best results, and can result in poor performance if production runs do not match profiling runs.



We considered but did not include conditional moves or predicated instructions, which can effectively replace unpredictable short forward branches. Conditional moves are the simpler of the two, but are difficult to use with conditional code that might cause exceptions (memory accesses and floating-point operations). Predication adds additional flag state to a system, additional instructions to set and clear flags, and additional encoding overhead on every instruction. Both conditional move and predicated instructions add complexity to out-of-order microarchitectures, adding an implicit third source operand due to the need to copy the original value of the destination architectural register into the renamed destination physical register if the predicate is false. Also, static compile-time decisions to use predication instead of branches can result in lower performance on inputs not included in the compiler training set, especially given that unpredictable branches are rare, and becoming rarer as branch prediction techniques improve.

We note that various microarchitectural techniques exist to dynamically convert unpredictable short forward branches into internally predicated code to avoid the cost of flushing pipelines on a branch mispredict (Heil & Smith, 1996), (Klauser et al., 1998), (Kim et al., 2005) and have been implemented in commercial processors (Sinharoy et al., 2011). The simplest techniques just reduce the penalty of recovering from a mispredicted short forward branch by only flushing instructions in the branch shadow instead of the entire fetch pipeline, or by fetching instructions from both sides using wide instruction fetch or idle instruction fetch slots. More

complex techniques for out-of-order cores add internal predicates on instructions in the branch shadow, with the internal predicate value written by the branch instruction, allowing the branch and following instructions to be executed speculatively and out-of-order with respect to other code.

The conditional branch instructions will generate an instruction-address-misaligned exception if the target address is not aligned to a four-byte boundary and the branch condition evaluates to true. If the branch condition evaluates to false, the instruction-address-misaligned exception will not be raised.



Instruction-address-misaligned exceptions are not possible on machines with IALIGN=16, e.g., those with the compressed instruction-set extension, C.

2.6. Load and Store Instructions

RV32I is a load-store architecture, where only load and store instructions access memory and arithmetic instructions only operate on CPU registers. RV32I provides a 32-bit address space that is byte-addressed. The EEI will define what portions of the address space are legal to access with which instructions (e.g., some addresses might be read only, or support word access only). Loads with a destination of `x0` must still raise any exceptions and cause any other side effects even though the load value is discarded.

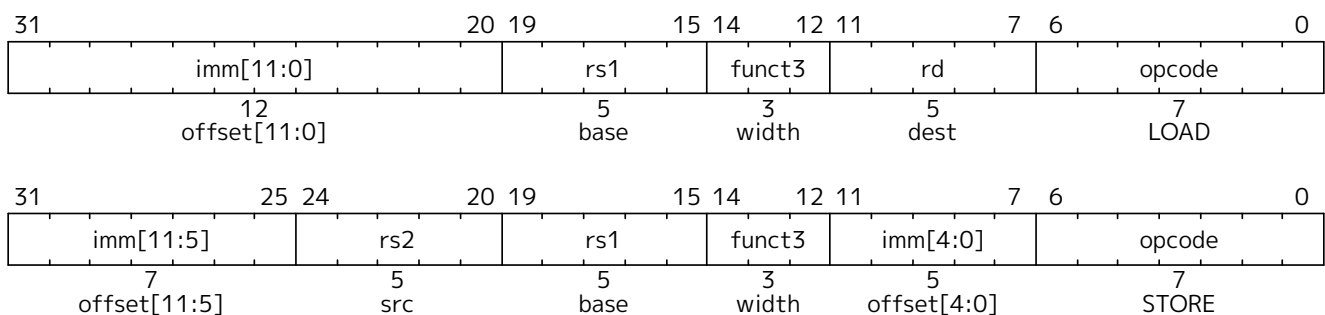
The EEI will define whether the memory system is little-endian or big-endian. In RISC-V, endianness is byte-address invariant.



In a system for which endianness is byte-address invariant, the following property holds: if a byte is stored to memory at some address in some endianness, then a byte-sized load from that address in any endianness returns the stored value.

In a little-endian configuration, multibyte stores write the least-significant register byte at the lowest memory byte address, followed by the other register bytes in ascending order of their significance. Loads similarly transfer the contents of the lesser memory byte addresses to the less-significant register bytes.

In a big-endian configuration, multibyte stores write the most-significant register byte at the lowest memory byte address, followed by the other register bytes in descending order of their significance. Loads similarly transfer the contents of the greater memory byte addresses to the less-significant register bytes.



Load and store instructions transfer a value between the registers and memory. Loads are encoded in the I-type format and stores are S-type. The effective address is obtained by adding register *rs1* to the sign-extended 12-bit offset. Loads copy a value from memory to register *rd*. Stores copy the value in register *rs2* to memory.

The LW instruction loads a 32-bit value from memory into *rd*. LH loads a 16-bit value from memory, then sign-extends to 32-bits before storing in *rd*. LHU loads a 16-bit value from memory but then zero extends to 32-bits before storing in *rd*. LB and LBU are defined analogously for 8-bit values. The SW, SH, and SB

instructions store 32-bit, 16-bit, and 8-bit values from the low bits of register *rs2* to memory.

Regardless of EEI, loads and stores whose effective addresses are naturally aligned shall not raise an address-misaligned exception. Loads and stores whose effective address is not naturally aligned to the referenced datatype (i.e., the effective address is not divisible by the size of the access in bytes) have behavior dependent on the EEI.

An EEI may guarantee that misaligned loads and stores are fully supported, and so the software running inside the execution environment will never experience a contained or fatal address-misaligned trap. In this case, the misaligned loads and stores can be handled in hardware, or via an invisible trap into the execution environment implementation, or possibly a combination of hardware and invisible trap depending on address.

An EEI may not guarantee misaligned loads and stores are handled invisibly. In this case, loads and stores that are not naturally aligned may either complete execution successfully or raise an exception. The exception raised can be either an address-misaligned exception or an access-fault exception. For a memory access that would otherwise be able to complete except for the misalignment, an access-fault exception can be raised instead of an address-misaligned exception if the misaligned access should not be emulated, e.g., if accesses to the memory region have side effects. When an EEI does not guarantee misaligned loads and stores are handled invisibly, the EEI must define if exceptions caused by address misalignment result in a contained trap (allowing software running inside the execution environment to handle the trap) or a fatal trap (terminating execution).



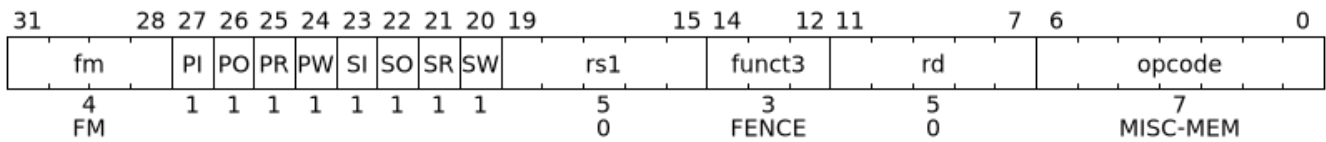
Misaligned accesses are occasionally required when porting legacy code, and help performance on applications when using any form of packed-SIMD extension or handling externally packed data structures. Our rationale for allowing EEIs to choose to support misaligned accesses via the regular load and store instructions is to simplify the addition of misaligned hardware support. One option would have been to disallow misaligned accesses in the base ISAs and then provide some separate ISA support for misaligned accesses, either special instructions to help software handle misaligned accesses or a new hardware addressing mode for misaligned accesses. Special instructions are difficult to use, complicate the ISA, and often add new processor state (e.g., SPARC VIS align address offset register) or complicate access to existing processor state (e.g., MIPS LWL/LWR partial register writes). In addition, for loop-oriented packed-SIMD code, the extra overhead when operands are misaligned motivates software to provide multiple forms of loop depending on operand alignment, which complicates code generation and adds to loop startup overhead. New misaligned hardware addressing modes take considerable space in the instruction encoding or require very simplified addressing modes (e.g., register indirect only).

Even when misaligned loads and stores complete successfully, these accesses might run extremely slowly depending on the implementation (e.g., when implemented via an invisible trap). Furthermore, whereas naturally aligned loads and stores are guaranteed to execute atomically, misaligned loads and stores might not, and hence require additional synchronization to ensure atomicity.



We do not mandate atomicity for misaligned accesses so execution environment implementations can use an invisible machine trap and a software handler to handle some or all misaligned accesses. If hardware misaligned support is provided, software can exploit this by simply using regular load and store instructions. Hardware can then automatically optimize accesses depending on whether runtime addresses are aligned.

2.7. Memory Ordering Instructions



FENCE instructions are used to order device I/O and memory accesses as viewed by other RISC-V harts and external devices or coprocessors. Any combination of device input (I), device output (O), memory reads (R), and memory writes (W) may be ordered with respect to any combination of the same. Informally, no other RISC-V hart or external device can observe any operation in the *successor* set following a FENCE before any operation in the *predecessor* set preceding the FENCE. [Chapter 18](#) provides a precise description of the RISC-V memory consistency model.

FENCE instructions also order memory reads and writes made by the hart as observed by memory reads and writes made by an external device. However, FENCE instructions do not order observations of events made by an external device using any other signaling mechanism.



A device might observe an access to a memory location via some external communication mechanism, e.g., a memory-mapped control register that drives an interrupt signal to an interrupt controller. This communication is outside the scope of the FENCE ordering mechanism and hence FENCE instructions can provide no guarantee on when a change in the interrupt signal is visible to the interrupt controller. Specific devices might provide additional ordering guarantees to reduce software overhead but those are outside the scope of the RISC-V memory model.

The EEI will define what I/O operations are possible, and in particular, which memory addresses when accessed by load and store instructions will be treated and ordered as device input and device output operations respectively rather than memory reads and writes. For example, memory-mapped I/O devices will typically be accessed with uncached loads and stores that are ordered using the I and O bits rather than the R and W bits. Instruction-set extensions might also describe new I/O instructions that will also be ordered using the I and O bits in a FENCE instruction.

Table 4. Fence mode encoding

<i>fm</i> field	Mnemonic suffix	Meaning
0000	<i>none</i>	Normal Fence
1000	.TSO	With FENCE RW,RW : exclude write-to-read ordering; otherwise: <i>Reserved for future use.</i>
<i>other</i>	<i>other</i>	<i>Reserved for future use.</i>

The FENCE mode field *fm* defines the semantics of the FENCE instruction. A **FENCE** (with *fm*=0000) orders all memory operations in its predecessor set before all memory operations in its successor set.

A **FENCE.TSO** instruction is encoded as a FENCE instruction with *fm*=1000, *predecessor*=RW, and *successor*=RW. **FENCE.TSO** orders all load operations in its predecessor set before all memory operations in its successor set, and all store operations in its predecessor set before all store operations in its successor set. This leaves **non-AMO** store operations in the **FENCE.TSO**'s predecessor set unordered with **non-AMO** loads in its successor set.



*Because **FENCE RW,RW** imposes a superset of the orderings that **FENCE.TSO** imposes, it is correct to ignore the *fm* field and implement **FENCE.TSO** as **FENCE RW,RW**.*

The unused fields in the FENCE instructions--*rs1* and *rd*--are reserved for finer-grain fences in future extensions. For forward compatibility, base implementations shall ignore these fields, and standard

software shall zero these fields. Likewise, many *fm* and predecessor/successor set settings are also reserved for future use. Base implementations shall treat all such reserved configurations as **FENCE** instructions (with *fm*=0000), and standard software shall use only non-reserved configurations.



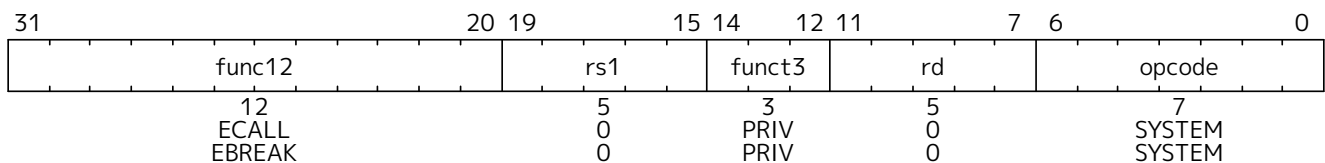
We chose a relaxed memory model to allow high performance from simple machine implementations and from likely future coprocessor or accelerator extensions. We separate out I/O ordering from memory R/W ordering to avoid unnecessary serialization within a device-driver hart and also to support alternative non-memory paths to control added coprocessors or I/O devices. Simple implementations may additionally ignore the predecessor and successor fields and always execute a conservative FENCE on all operations.

2.8. Environment Call and Breakpoints

SYSTEM instructions are used to access system functionality that might require privileged access and are encoded using the I-type instruction format. These can be divided into two main classes: those that atomically read-modify-write control and status registers (CSRs), and all other potentially privileged instructions. CSR instructions are described in [Chapter 6](#), and the base unprivileged instructions are described in the following section.



*The **SYSTEM** instructions are defined to allow simpler implementations to always trap to a single software trap handler. More sophisticated implementations might execute more of each system instruction in hardware.*



These two instructions cause a precise requested trap to the supporting execution environment.

The **ECALL** instruction is used to make a service request to the execution environment. The **EEL** will define how parameters for the service request are passed, but usually these will be in defined locations in the integer register file.

The **EBREAK** instruction is used to return control to a debugging environment.



ECALL and EBREAK were previously named SCALL and SBREAK. The instructions have the same functionality and encoding, but were renamed to reflect that they can be used more generally than to call a supervisor-level operating system or debugger.

EBREAK was primarily designed to be used by a debugger to cause execution to stop and fall back into the debugger. EBREAK is also used by the standard GCC compiler to mark code paths that should not be executed.



Another use of EBREAK is to support "semlhosting", where the execution environment includes a debugger that can provide services over an alternate system call interface built around the EBREAK instruction. Because the RISC-V base ISAs do not provide more than one EBREAK instruction, RISC-V semlhosting uses a special sequence of instructions to distinguish a semlhosting EBREAK from a debugger inserted EBREAK.

```
slli x0, x0, 0x1f    # Entry NOP
ebreak               # Break to debugger
```

```
srai x0, x0, 7      # Exit NOP
```

Note that these three instructions must be 32-bit-wide instructions, i.e., they mustn't be among the compressed 16-bit instructions described in [Chapter 28](#).

The shift NOP instructions are still considered available for use as HINTs.

Semihosting is a form of service call and would be more naturally encoded as an ECALL using an existing ABI, but this would require the debugger to be able to intercept ECALLs, which is a newer addition to the debug standard. We intend to move over to using ECALLs with a standard ABI, in which case, semihosting can share a service ABI with an existing standard.

We note that ARM processors have also moved to using SVC instead of BKPT for semihosting calls in newer designs.

2.9. HINT Instructions

RV32I reserves a large encoding space for HINT instructions, which are usually used to communicate performance hints to the microarchitecture. Like the NOP instruction, HINTs do not change any architecturally visible state, except for advancing the **pc** and any applicable performance counters. Implementations are always allowed to ignore the encoded hints.

Most RV32I HINTs are encoded as integer computational instructions with $rd=x0$. The other RV32I HINTs are encoded as FENCE instructions with a null predecessor or successor set and with $fm=0$.

These HINT encodings have been chosen so that simple implementations can ignore HINTs altogether, and instead execute a HINT as a regular instruction that happens not to mutate the architectural state. For example, ADD is a HINT if the destination register is **x0**; the five-bit *rs1* and *rs2* fields encode arguments to the HINT. However, a simple implementation can simply execute the HINT as an ADD of *rs1* and *rs2* that writes **x0**, which has no architecturally visible effect.



As another example, a FENCE instruction with a zero *pred* field and a zero *fm* field is a HINT; the *succ*, *rs1*, and *rd* fields encode the arguments to the HINT. A simple implementation can simply execute the HINT as a FENCE that orders the null set of prior memory accesses before whichever subsequent memory accesses are encoded in the *succ* field. Since the intersection of the predecessor and successor sets is null, the instruction imposes no memory orderings, and so it has no architecturally visible effect.

[Table 5](#) lists all RV32I HINT code points. 91% of the HINT space is reserved for standard HINTs. The remainder of the HINT space is designated for custom HINTs: no standard HINTs will ever be defined in this subspace.



We anticipate standard hints to eventually include memory-system spatial and temporal locality hints, branch prediction hints, thread-scheduling hints, security tags, and instrumentation flags for simulation/emulation.

Table 5. RV32I HINT instructions.

Instruction	Constraints	Code Points	Purpose
LUI	$rd = x0$	2^{20}	<i>Designated for future standard use</i>
AUIPC	$rd = x0$	2^{20}	
ADDI	$rd = x0$, and either $rs1 \neq x0$ or $imm \neq 0$	$2^{17} - 1$	
ANDI	$rd = x0$	2^{17}	
ORI	$rd = x0$	2^{17}	
XORI	$rd = x0$	2^{17}	
ADD	$rd = x0, rs1 \neq x0$	$2^{10} - 32$	
ADD	$rd = x0, rs1 = x0, rs2 \neq x2 - x5$	28	
ADD	$rd = x0, rs1 = x0, rs2 = x2 - x5$	4	($rs2 = x2$) NTL.P1 ($rs2 = x3$) NTL.PALL ($rs2 = x4$) NTL.S1 ($rs2 = x5$) NTL.ALL
SLLI	$rd = x0, rs1 = x0, shamt = 31$	1	Semihosting entry marker
SRAI	$rd = x0, rs1 = x0, shamt = 7$	1	Semihosting exit marker
SUB	$rd = x0$	2^{10}	<i>Designated for future standard use</i>
AND	$rd = x0$	2^{10}	
OR	$rd = x0$	2^{10}	
XOR	$rd = x0$	2^{10}	
SLL	$rd = x0$	2^{10}	
SRL	$rd = x0$	2^{10}	
SRA	$rd = x0$	2^{10}	
FENCE	$rd = x0, rs1 \neq x0, fm = 0$, and either $pred = 0$ or $succ = 0$	$2^{10} - 63$	
FENCE	$rd \neq x0, rs1 = x0, fm = 0$, and either $pred = 0$ or $succ = 0$	$2^{10} - 63$	
FENCE	$rd = rs1 = x0, fm = 0, pred = 0, succ \neq 0$	15	
FENCE	$rd = rs1 = x0, fm = 0, pred \neq W, succ = 0$	15	
FENCE	$rd = rs1 = x0, fm = 0, pred = W, succ = 0$	1	PAUSE
SLTI	$rd = x0$	2^{17}	<i>Designated for custom use</i>
SLTIU	$rd = x0$	2^{17}	
SLLI	$rd = x0$, and either $rs1 \neq x0$ or $shamt \neq 31$	$2^{10} - 1$	
SRLI	$rd = x0$	2^{10}	
SRAI	$rd = x0$, and either $rs1 \neq x0$ or $shamt \neq 7$	$2^{10} - 1$	
SLT	$rd = x0$	2^{10}	
SLTU	$rd = x0$	2^{10}	



`slli x0, x0, 0x1f` and `srai x0, x0, 7` were previously designated as custom HINTs, but they have been appropriated for use in semihosting calls, as described in [Section 2.8](#). To reflect their usage in practice, the base ISA spec has been changed to designate them as standard HINTs.

Chapter 3. RV32E and RV64E Base Integer Instruction Sets, Version 2.0

This chapter describes the RV32E and RV64E base integer instruction sets, designed for microcontrollers in embedded systems. RV32E and RV64E are reduced versions of RV32I and RV64I, respectively: the only change is to reduce the number of integer registers to 16. This chapter only outlines the differences between RV32E/RV64E and RV32I/RV64I, and so should be read after [Chapter 2](#) and [Chapter 4](#).



RV32E was designed to provide an even smaller base core for embedded microcontrollers. There is also interest in RV64E for microcontrollers within large SoC designs, and to reduce context state for highly threaded 64-bit processors.

Unless otherwise stated, standard extensions compatible with RV32I and RV64I are also compatible with RV32E and RV64E, respectively.

3.1. RV32E and RV64E Programmers' Model

RV32E and RV64E reduce the integer register count to 16 general-purpose registers, (**x0-x15**), where **x0** is a dedicated zero register.



We have found that in the small RV32I core implementations, the upper 16 registers consume around one quarter of the total area of the core excluding memories, thus their removal saves around 25% core area with a corresponding core power reduction.

3.2. RV32E and RV64E Instruction Set Encoding

RV32E and RV64E use the same instruction-set encoding as RV32I and RV64I respectively, except that only registers **x0-x15** are provided. All encodings specifying the other registers **x16-x31** are reserved.



*The previous draft of this chapter made all encodings using the **x16-x31** registers available as custom. This version takes a more conservative approach, making these reserved so that they can be allocated between custom space or new standard encodings at a later date.*